Final Report (LANA)

Project Title: Understanding Financial Market Dynamics through Linear Algebra

1. Student Information

- **Name:** Vicky Patel
- **Student ID:** B24BS2412
- **Batch/Section: BSc/ BS in applied Data Science and AI (**Cohort-2, Semester 1)
- **Date of Submission: Sunday, 12th October 2025**
- **Course: Linear Algebra and Numerical Analysis**

---

2. Project Overview

- **Objective of the Project:**

  The main objective of the project is to use Python, Linear Algebra, and real-world data to analyze how shocks in the market affect the stability of these financial institutions in India, such as SBI, ICICI, HDFC, and Axis Bank. We have to calculate Eigenvalues , Eigenvectors, etc in the project for the analysis.

- **Tools Used:**

  Google Colab (Library used: **Numpy, Sympy, matplotlib)** for generating output for the given tasks and also used for graph visualization. Used **Google Docs** for final report , recommendations , and interpret insights and visualizations.

🧮 3. Task 1 – Matrix Analysis

🔹 **Objective:** Understand the basics of matrices financial modelling.

Given:

$$A = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \end{bmatrix}$$

Initial portfolio vector:

$$x = \begin{vmatrix} 1 \\ 1 \\ 0 \\ 0 \end{vmatrix}$$

- **Findings:**

1. Matrix A (representing connections between banks) is **symmetric**.
2. We got rank **4** of matrix A that is full rank.
3. The determinant of the Matrix A is **4.9999 ~ 5** that is non zero means the matrix A is invertible, the system has a unique and stable solution.
4. While calculating the trace of matrix A we got value **8** which means total-self dependency of banks.
5. Null Space: only trivial solution **(X = 0)** that shows no redundant relationships.
6. Solving the system of equations (AX = B) we got **[3, 5, 5, 3]**.

Colab Output Screenshot:

```
Symmetric: True
Rank: 4
Determinant: 4.999999999999999
Trace: 8
Null Space:
Only trivial solution (x = 0)
Solution: [3. 5. 5. 3.]
Ax = b: [1. 2. 2. 1.]
```

Interpretation:

- Matrix A is symmetric and full rank showing a **well-connected yet stable** network of banks.
- det(A) not equal to 0 means small changes in one bank's condition **will not collapse the entire system**, indicating **mathematical and financial stability**.
- The Rank is 4 (full rank) that shows independent relationships between banks, we **cannot derive financial patterns** for any banks from a combination of others.
- Null Space is 0 shows **no hidden dependency** in the system, each bank plays a distinct role in maintaining balance.
- The solution of the system of equations is in equilibrium state that means **ICICI and HDFC are more sensitive** to market changes whose **value is 5** and **SBI and Axis Bank (value 3)** remain relatively stable.

⚙️ 4. Task 2 – Eigenvalue & Eigenvector Analysis

🔹 **Objective:** Explore how market shocks propagate through the interconnected financial network.

🔹 **Findings:**

1. Eigenvalues: **[0.382, 1.382, 2.618, 3.618]** (rounded up to nearest integer).
2. Eigenvectors:
   [[-0.372 -0.601 -0.372 -0.602]
   [0.602 0.372 -0.602 -0.372]
   [-0.602 0.372 -0.602 0.372]
   [0.372 -0.602 -0.372 0.602]]
3. Largest (Dominant) and smallest (vulnerable) eigenvalues are **3.618** and **0.381** respectively.
4. Cayley–Hamilton Theorem: verified **p(A) = 0**, which confirms that A satisfies its own characteristic equation.

<u>Colab Output Screenshot:</u>

```
Eigenvalues: [3.61803399 2.61803399 0.38196601 1.38196601]
Eigenvectors: [[-0.37174803 -0.60150096 -0.37174803 -0.60150096]
 [ 0.60150096  0.37174803 -0.60150096 -0.37174803]
 [-0.60150096  0.37174803 -0.60150096  0.37174803]
 [ 0.37174803 -0.60150096 -0.37174803  0.60150096]]
Dominant (largest) eigenvalues: 3.6180339887498913
Vulnearable (smallest) eigenvalues: 0.3819660112501053
Cayley-Hamilton Validation-- (should be near zero matrix):
 [[ 2.84217094e-14 -2.13162821e-14  7.10542736e-15 -2.66453526e-15]
 [-2.13162821e-14  2.84217094e-14 -2.13162821e-14  7.10542736e-15]
 [ 7.10542736e-15 -2.13162821e-14  2.84217094e-14 -2.13162821e-14]
 [-2.66453526e-15  7.10542736e-15 -2.13162821e-14  2.84217094e-14]]
```

<u>Interpretation:</u>

- Here all eigenvalues are **positive** which shows a stable system means that the overall financial system is **mathematically stable**.
- The dominant Eigenvalue 3.618 indicates that **certain combinations of banks** are highly sensitive to changes — small market shocks can create **large effects** in those directions. And smallest eigenvalues represent the most stable pattern.
- Each eigenvector shows how the banks move **together or oppositely** during market fluctuations.
- By validating Cayley-Hamilton Theorem we can confirm that the system is **mathematically validating the network structure**.

📘 5. Task 3 – Portfolio Risk Assessment

🔹 **Objective:** To Simulate how stock prices evolve under market shocks and understand risk in financial portfolios.

🔹 **Step 1 – Portfolio Span Check**

🔹 **Findings:**

1. Portfolio vector is **[1 1 0 0]** , initial state of portfolio.

2. Projections on dominant eigenvectors are **[0.0528 0.0528 -0.2236 0.2236]**, this shows the Portfolio component in risky directions.

3. Reconstruction error is **1.376**. Non-zero means portfolio nor exactly in the dominant eigenvector span.

Colab Output Screenshot:

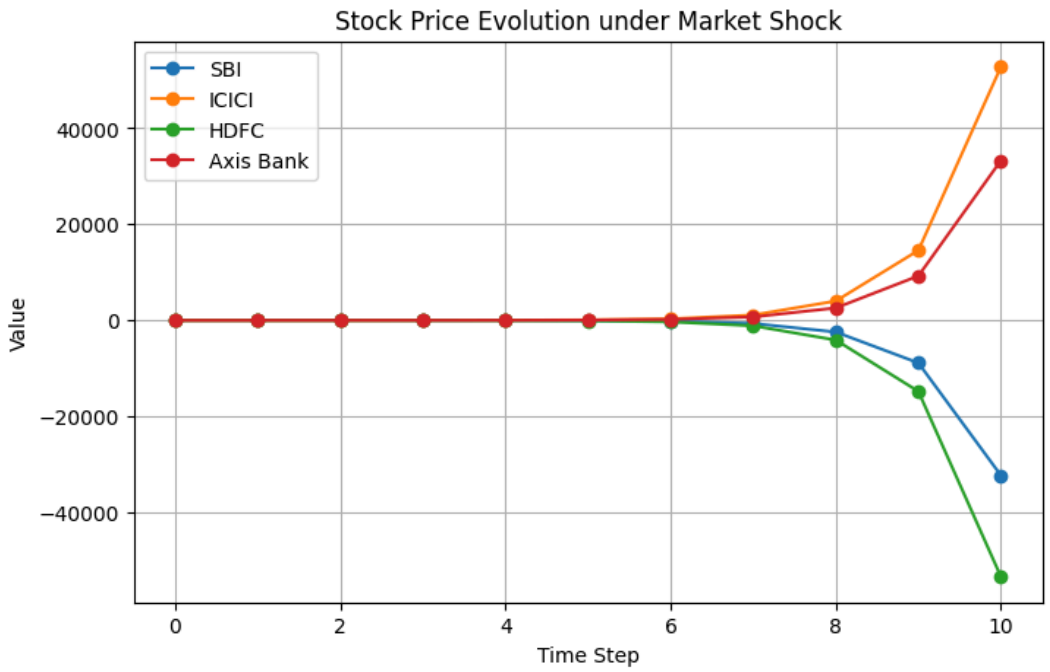
```
=== Step 1: Portfolio span check ===
Portfolio vector: [1. 1. 0. 0.]
Projection onto dominant eigenvectors: [ 0.0527864  0.0527864 -0.2236068  0.2236068]
Reconstruction error: 1.3763819204711738
Result: Portfolio does NOT exactly lie in the span (approx only)
```

Interpretation:

- From our findings this can be seen that the portfolio is partially aligned with risky modes but not completely controlled by them. It means that it has moderate exposure to shock-amplifying directions.

🔹 **Step 2 – Market Shock Simulation**

Obtained Graph From Colab:



1. **ICICI (orange)** → shows sharp, exponential-like growth after step 7 → **most shock-sensitive**.

2. **Axis Bank (red)** → also increases steadily but slower → **moderate sensitivity**.

3. **SBI (blue)** and **HDFC (green)** → decline rapidly after step 7 → **negatively affected**, experiencing losses.

4. The values diverge strongly by step 10 → the system becomes **unstable under continuous shocks**.

Overall Findings and Interpretation:

1. Portfolio not in dominant span that means moderate but real exposure to **unstable market directions**.

2. Divergence in simulation shows that the system is unstable under repeated shocks (**largest eigenvalue > 1**).

3. ICICI amplifies shocks that a **highly sensitive bank needs stronger control**.

4. Axis bank stable side means that it acts as **partial stabilizer**.

5. SBI and HDFC decline means **Vulnerable to negative propagation**.

6. Overall trend shows uneven and **asymmetric propagation of risk**.

📈 Conclusion & Recommendations

- **Conclusion:**
  The dynamic model shows that the financial network becomes unstable when exposed to continuous shocks. The divergence between banks proves that **interconnectedness can amplify instability** even in a mathematically symmetric system.

- **Recommendations:**

  1. **Diversify** portfolios to reduce exposure to dominant eigenmodes.

2. **Apply damping or regulation** (e.g., reduce interbank influence coefficients).

3. **Stress-test** regularly by observing changes in dominant eigenvalues.

4. **Provide capital buffers** for highly sensitive banks like ICICI.

📘 6. Task 4 – Storytelling with Data

🔹 **Objective:** To present a creative solution for stabilizing the financial network during economic turbulence from the insights of the previous task analyses.

What We Found Earlier:

- **Task 1:** The matrix A was **symmetric and invertible**, meaning the network of banks is connected and mathematically stable.

- **Task 2:** All eigenvalues were positive, confirming stability, but the **largest eigenvalue (3.618)** showed the direction where shocks can grow very fast.

- **Task 3:** The simulation proved that **ICICI amplifies shocks the most**, SBI and HDFC are negatively affected, and Axis Bank stays relatively stable.

The Story Behind the Data:

1. ICICI's value kept rising sharply — it **amplified** the shock.
2. SBI and HDFC's values dropped — they **absorb losses**.
3. Axis Bank increased slowly — it's **comparatively stable**.

Interpretation:

- From findings we can say that the system is mathematically stable and repeated shocks make it unstable over time.
- ICICI amplifies shocks the most, while SBI and HDFC show negative effects.
- This shows how systemic risk can develop in a connected financial system.

Recommendations:

1. **Maintain capital buffers** for the sensitive banks like ICICI and should keep extra reserves.

2. **Monitor the dominant eigenvalue**, if it increases the system will become more fragile.
3. **Reduce interbank connections** for weaker connections that limit shock spreading.
4. **Diversify** portfolios to reduce exposure to dominant eigenmodes.

Conclusion:

- From this capstone project we learned that linear algebra is not just theoretical math, it can help explain how real financial systems behave.
- We calculated and analysed eigenvalues in which we discovered how dominant eigenvalues show market shocks.
- We observed from the graph that ICICI amplifies shocks, while Axis Bank seems more stable.

End Of Report